Module7_Lab3

October 17, 2026

[1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt

df = pd.read_csv('titanic_train.csv')

features = ['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']
X = df[features].copy()
y = df['Survived']

X['Age'] = X['Age'].fillna(X['Age'].median())
X['Embarked'] = X['Embarked'].fillna(X['Embarked'].mode()[0])

X = pd.get_dummies(X, columns=['Sex', 'Embarked'], drop_first=True)

X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2,␣
 ↪random_state=42)

dt_clf = DecisionTreeClassifier(max_depth=3, random_state=42)
dt_clf.fit(X_train, y_train)

y_pred = dt_clf.predict(X_val)
accuracy = accuracy_score(y_val, y_pred)

print(f"Validation Accuracy: {accuracy:.4f}\n")
print("Classification Report:")
print(classification_report(y_val, y_pred))

plt.figure(figsize=(12, 6))
plot_tree(
    dt_clf,
    feature_names=X.columns.tolist(),
    class_names=['Not Survived', 'Survived'],
    filled=True,
    rounded=True
```

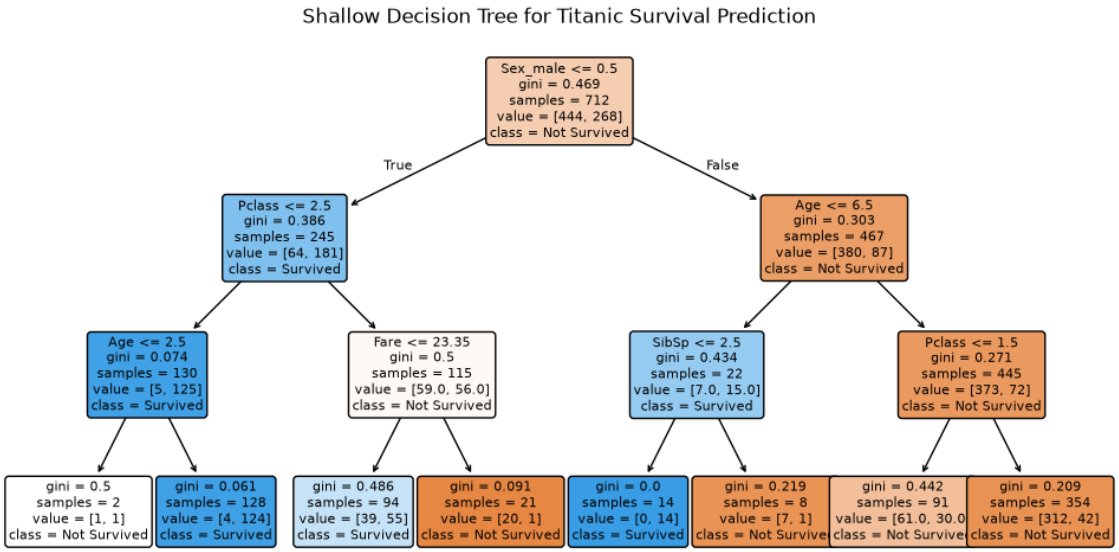
```
)
plt.title("Shallow Decision Tree for Titanic Survival Prediction")
plt.show()
```

```
Validation Accuracy: 0.7989

Classification Report:
              precision    recall  f1-score   support

           0       0.80      0.88      0.84       105
           1       0.80      0.69      0.74        74

    accuracy                           0.80       179
   macro avg       0.80      0.78      0.79       179
weighted avg       0.80      0.80      0.80       179
```



I picked Pclass, sex, age, and fare. To maintain an honest evaluation, data was split before imputing missing age values, so statistics from the test set don't leak in the training process.

[ ]: